

Arithmetic and complexity

Realizing every sign-matrix permanent with negative entries confined on or above the diagonal

Difficulty: challenging **Importance:** interesting to specialist

Status: Open

Rating rationale: Challenging because equality of the full permanent ranges requires a uniform value-preserving construction or an obstruction; specialist importance concerns structural realization of discrete matrix invariants.

Problem statement

For each integer $n \geq 1$, let

$$\Omega_n = \{-1, 1\}^{n \times n}, \quad \mathcal{U}_n = \{B \in \Omega_n : b_{ij} = 1 \text{ whenever } i > j\}.$$

Define $\text{per } A = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}$. Does every $A \in \Omega_n$ admit some $B \in \mathcal{U}_n$ with

$$\text{per } B = \text{per } A?$$

Equivalently, do Ω_n and $\mathcal{U}_n$ have identical permanent ranges for every n ? Entries below the diagonal are $+1$, not zero. Negative entries are allowed on the diagonal.

Relevance and ratings

A positive answer would reduce exact value realization to a smaller, explicitly structured family of matrices. The question connects matrix structure and the combinatorial complexity of the permanent; it does not ask for an efficient permanent-evaluation algorithm. The main significance is to specialists, and a uniform structural argument or counterexample is required.

References

- D. Ingram and A. Razborov, *On the range of the permanent of (± 1) -matrices*, Linear Algebra Appl. 743 (2026), 271–285 ([journal](#)); [arXiv:2507.09433v1](#), §6, Problem 4 and the paragraph defining the restricted family.
- Z. Hunter, M. Kwan and L. Sauermann, *Exponential anticoncentration of the permanent* (2025), §1.2, Corollary 1.3 ([primary paper](#)).

Status check — 2026-09-10

Checked [Ingram–Razborov, §6, Problem 4](#), including the preceding definition: negative entries may occur on or above the diagonal. The displayed target matches that question. Exact-title and upper-triangular permanent-range searches found no general resolution. [Hunter–Kwan–Sauermann, Corollary 1.3](#), settles exponential growth of the unrestricted range, not equality with this restricted family; rank-dependent maximum results also leave exact ranges undetermined.